

HW Bayes

a) The maximum a posteriori knowledge (MAP) hypothesis can be found by

$$\text{MAP}(q) = \underset{c \in C}{\text{argmax}} P(c).$$

If $C = \{c_1, c_2\}$ is dichotomy, i.e. $|C| = 2$.

$$\text{MAP}(q) = \begin{cases} c_1 & \text{if } P(c_1) > P(c_2) \\ c_2 & \text{otherwise} \end{cases}$$

Since $P(V) = 0.02$ and $P(\bar{V}) = 0.98$, the person is not infected with COVID-19. Note that $\bar{V}$ is the event that a person in the state is not infected with COVID-19.

b) The maximum likelihood (ML) hypothesis given the evidence E can be found by

$$\text{ML}(q) = \underset{c \in C}{\text{argmax}} P(E|c)$$

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If $C = \{c_1, c_2\}$ is dichotomy, i.e. $|C| = 2$.

$$\text{ML}(q) = \begin{cases} c_1 & \text{if } P(E|c_1) > P(E|c_2) \\ c_2 & \text{otherwise} \end{cases}$$

From the sentence "a test kit A which has a 3% chance of showing positive on a normal person" $P(A^+/\bar{V}) = 0.03$

From the sentence "a 2% chance showing negative on an infected person"

$P(A^-/V) = 0.02$, and thus $P(A^+/V) = 0.98$

Since $[P(A^+/V) = 0.98] > [P(A^+/\bar{V}) = 0.03]$, the person is infected with COVID-19.

- c) The maximum likelihood (ML) hypothesis given the evidence E can be found by

$$ML(q) = \underset{c \in C}{\text{argmax}} P(E/c)$$

If $C = \{c_1, c_2\}$ is dichotomy, i.e. $|C| = 2$,

$$ML(q) = \begin{cases} c_1 & \text{if } P(E/c_1) > P(E/c_2) \\ c_2 & \text{otherwise} \end{cases}$$

From the sentence "a test kit A which has 3% chance of showing positive on a normal person" $P(A^+/V) = 0.03$.

From the sentence "a 2% chance showing negative on an infected person" $P(A^-/V) = 0.02$, and thus $P(A^+/V) = 0.98$

Since $[P(A^+/V) = 0.98] > [P(A^+/\bar{V}) = 0.03]$, the person is infected with COVID-19.

$$\begin{aligned} d) P(V/A^+) &= \frac{P(A^+/V)P(V)}{P(A^+)} \\ &= \frac{P(A^+/V)P(V)}{P(A^+/V)P(V) + P(A^+/\bar{V})P(\bar{V})} \end{aligned}$$

$$= \frac{0.02}{0.02 + 0.03} = \frac{0.02}{0.05}$$

$$= 0.4.$$

- e) Since the test is conducted independently, it can be assumed that

$$P(A^+, A^+/V) = P(A^+/V) P(A^+/W)$$

$$\text{Since } [P(A^+/V) P(A^+/V) P(V) = 0.019208] >$$

$$[P(A^+/V) P(A^+/V) P(V) = 0.000882]$$

the person is infected with COVID-19.

Q1.2

- a) To find: $P(C_{19}/C_g=Y)$

$$P(C_{19}/C_g=Y) = \frac{0.4 \times 0.1}{0.4 \times 0.1 + 0.3 \times 0.9} = \frac{0.04}{0.04 + 0.27}$$

$$= \frac{0.04}{0.31} = 0.12903$$

- b) $P(C_g=Y, C_o=Y, D_i=N, F_a=Y, F_c=N, M_a=Y, S_t=N/C_{19}) =$

$$P(C_g=Y/C_{19}) \times P(C_o=Y/C_{19}) \times P(D_i=N/C_{19}) \times P(F_a=Y/C_{19}) \times$$

$$P(F_c=N/C_{19}) \times P(M_a=Y/C_{19}) \times P(S_t=N/C_{19})$$

$$= 0.4 \times 0.3 \times 0.65 \times 0.3 \times 0.6 \times 0.5 \times 0.55$$

$$= 0.003861$$

- c) $P(C_g=Y, C_o=Y, \overset{D_i=N}{F_a=Y}, F_c=N, M_a=Y, S_t=N/\overset{C_{19}}{\bar{C}_{19}}) =$

$$P(C_g=Y/\bar{C}_{19}) \times P(C_o=Y/\bar{C}_{19}) \times P(F_a=Y/\bar{C}_{19}) \times P(D_i=N/\bar{C}_{19}) \times$$

$$P(F_c=N/\bar{C}_{19}) \times P(M_a=Y/\bar{C}_{19}) \times P(S_t=N/\bar{C}_{19})$$

$$= 0.3 \times 0.2 \times 0.1 \times 0.9 \times 0.9 \times 0.1 \times 0.9 = 0.0004374$$

Infected because

$$(P(C_g=Y, C_o=Y, D_i=N, F_a=Y, F_c=N, M_a=Y, St=N / C_{19}) = 0.003861) >$$

$$(P(C_g=Y, C_o=Y, D_i=N, F_a=Y, F_c=N, M_a=Y, St=N / \bar{C}_{19}) = 0.0004374).$$

$$d) P(C_{19} / C_g=Y, C_o=Y, D_i=N, F_a=Y, F_c=N, M_a=Y, St=N)$$

$$= \frac{0.1 \times 0.003861}{0.1 \times 0.003861 + 0.9 \times 0.0004374}$$

$$= 0.49575$$

e) Not infected because

$$(P(C_{19} / C_g=Y, C_o=Y, D_i=N, F_a=Y, F_c=N, M_a=Y, St=N) = 0.49575) <$$

$$(P(\bar{C}_{19} / C_g=Y, C_o=Y, D_i=N, F_a=Y, F_c=N, M_a=Y, St=N) = 0.50485).$$

Q1.3.

a) Find the priori probability of French in the database, i.e. $P(F)$

$$P(F) = \frac{4}{20} = \frac{1}{5} = 0.2.$$

b) Find the conditional probabilities of $P(A/S)$ and $P(\bar{A}/F)$

$$P(A/S) = \frac{2}{3} \approx 0.6667 \text{ and } P(\bar{A}/F) = \frac{1}{2} = 0.5$$

c) $MP12(q) = \underset{C \in \{F, S\}}{\text{argmax}} P(C)$

English because it has the highest priori probability.

- d) The naive likelihood classifier (NLC) given the evidence $q = (1, 1, 0, 1)$ can be found by

$$\begin{aligned} \text{NLC}(q) &= \underset{c \in C}{\text{argmax}} P(a_1, \dots, a_d | c) \\ &= \underset{c \in C}{\text{argmax}} \prod_{i=1}^d P(a_i | c) \end{aligned}$$

Note that $a_1 = A$, $a_2 = B$, $a_3 = C$, $a_4 = D$.

$$\begin{aligned} \text{For English, } P(A, B, C, D | E) &= P(A | E) P(B | E) P(C | E) P(D | E) \\ &= \frac{2}{5} \times \frac{4}{5} \times \frac{1}{10} \times \frac{1}{5} \\ &= 0.0064. \end{aligned}$$

$$\begin{aligned} \text{For French, } P(A, B, C, D | F) &= P(A | F) P(B | F) P(C | F) P(D | F) \\ &= \frac{1}{2} \times \frac{3}{4} \times \frac{1}{1} \times \frac{1}{2} \\ &= 0.1875 \end{aligned}$$

$$\begin{aligned} \text{For Spanish, } P(A, B, C, D | S) &= P(A | S) P(B | S) P(C | S) P(D | S) \\ &= \frac{2}{3} \times \frac{5}{6} \times \frac{1}{6} \times \frac{1}{6} \\ &= 0.015432. \end{aligned}$$

- e) The naive Bayes classifier (NBC) given the evidence $q = (1, 1, 0, 1)$ can be found by
- $$\text{NBC}(q) = \underset{c \in C}{\text{argmax}} P(c) P(a_1, \dots, a_d | c) = \underset{c \in C}{\text{argmax}} P(c) \prod_{i=1}^d P(a_i | c)$$

Note that $a_1 = A$, $a_2 = B$, $a_3 = C$, $a_4 = D$.

$$\begin{aligned} \text{For English, } P(E) P(A | E) P(B | E) P(C | E) P(D | E) &= \frac{1}{2} \times \frac{2}{5} \times \frac{4}{5} \times \frac{1}{10} \times \frac{1}{5} = 0.0032 \\ \text{For French, } P(F) P(A | F) P(B | F) P(C | F) P(D | F) &= \frac{1}{3} \times \frac{1}{2} \times \frac{3}{4} \times \frac{1}{1} \times \frac{1}{2} = 0.0375 \\ \text{For Spanish, } P(S) P(A | S) P(B | S) P(C | S) P(D | S) &= \frac{2}{10} \times \frac{2}{3} \times \frac{5}{6} \times \frac{1}{6} \times \frac{1}{6} = 0.0046296. \end{aligned}$$